YouTube-DNN-RecSys — Candidate Generation Architecture

Data processing, storage, dnn.py model, prediction services, and support utilities

Dashed frame = architecture zone · Solid frame = subsystem · Box = module and responsibility

PROJECT OVERVIEW

YouTube-DNN-RecSys is a TensorFlow and Spark implementation of the candidate-generation stage of YouTube recommendations.
The architecture comprises data processing, storage, the dnn.py model, prediction services, and support utilities.

DATA PROCESSING · offline sample build

Python Pipeline

data2tfrecords.py
random training data

TFRecords
train / evaluation

read_tfrecords.py
dataset reader

Spark Pipeline

MakeDataOne.scala
distributed generation

MakeDataTwo.scala
TFRecords output

HDFS
dataset storage

STORAGE · datasets and checkpoints

data/tfrecords/
train/train.tfrecords
evaluation/evaluation.tfrecords

data/checkpoints/
ckpt/checkpoint
ckpt/eval · modelpath/

MODEL TRAINING · dnn.py

Input
visit_items_index × 5
continuous_features × 16

Item Embedding
500 items × 16

Average Pooling
history vector

DNN Tower
Dense 64 + ReLU
Dropout · Dense 32 + ReLU
user vector 16

Softmax Output
500 classes
cross entropy · Top-K

PREDICTION · serving paths

PredictUserVector.scala
SavedModel
user vector head

ItemEmbeddingPredictor.scala
item embedding table
FeatureBuilder

load_dnn_model.py
SavedModel load
online / batch inference

UTILITIES · training support

tensor_board.py
TensorBoard

feature_column.py
feature spec

self_defined_network_layer.py
custom TF layers

Spark Base
BaseSparkLocal.scala
BaseSparkOnline.scala